Project Title: Monitoring Environmental Conditions Using a Simplified DIY Sensor.

Do-it-yourself (DIY) sensors are low-cost sensors built around variables that one wants to monitor. In this project, I built an environmental system to monitor and log temperature, relative humidity, and carbon dioxide in my office. The system is composed of an Arduino Uno microcontroller serving as the 'brain.' Temperature/relative humidity sensor, carbon dioxide sensor, and SD card for logging and storing the data. A detailed description of the components is presented in the table below.

Table 1: The components used to build the system, including the name of the company that provides, the source link, and the current price found on their website

Component	Company	Source link	Comments	Current price
Arduino Uno R3	Arduino	https://store.arduino.cc/products/arduino-uno-rev3/	Controls the functionality of the sensors	€29.30
Adafruit SCD -30- NDIR True CO2 Temperature and Humidity Sensor	Adafruit	https://www.adafruit.com/product/4867#description	Measures real CO2 and has embedded T/RH	$ 58.95
Adafruit Sensirion SHT40 Temperature & Humidity Sensor	Adafruit	https://www.adafruit.com/product/4885	Measures T/RH with ± 0.2 ºC and ±1.8 %RH accuracy, respectively	$ 5.95
MicroSD card breakout board	Adafruit	https://www.adafruit.com/product/254	SPI interface with VCC, CS, D1, SCK, DO, CD, and GND pins	$ 7.50
32 GB SanDisk Ultra microSD card				

System Design

The experimental design involved soldering the SHT40 sensor and connecting the wires to the respective pins of the Audino Uno board. Similarly, soldering was completed for both the SDD-30 and MicroSD card breakout boards, and they were connected to the Uno board. A splitter was used to break the power from the 3.3V microcontroller board into two, so that the SH4T0 and SCD-30 CO2 sensors could tap power from each pin of the splitter. Pictures of the system design are shown below:

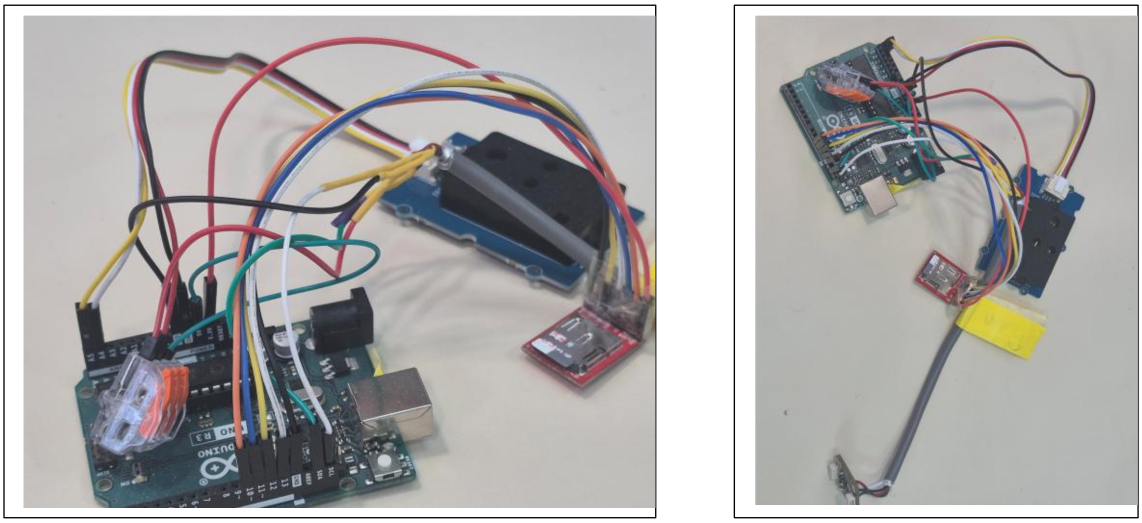


Figure 1 : Complete System of CO2, T/RH DIY sensor

Connection Diagram

The connection diagram shows the connections in the system. The Fritzing software was used for constructing the diagram, as presented below.

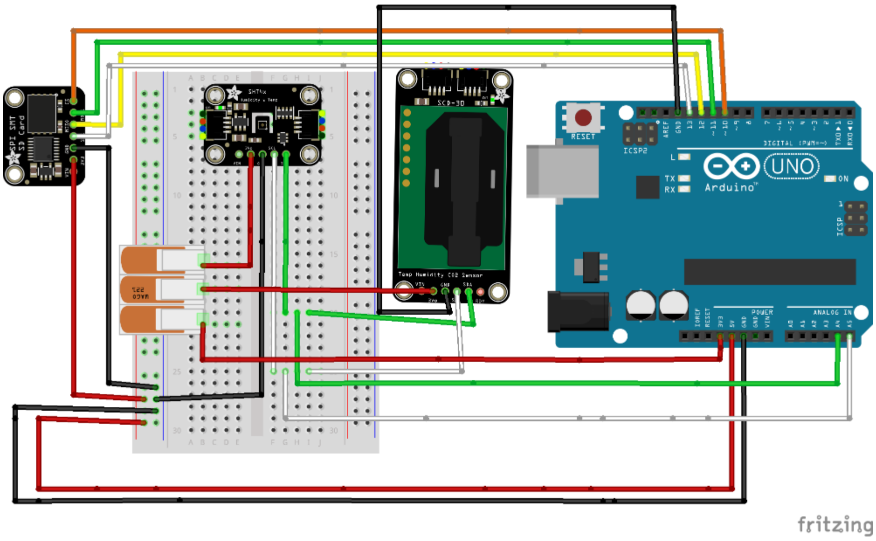


Figure 2 : Connection diagram of the Arduino Uno, SHT40, SCD30 sensors, and MicroSD breakout

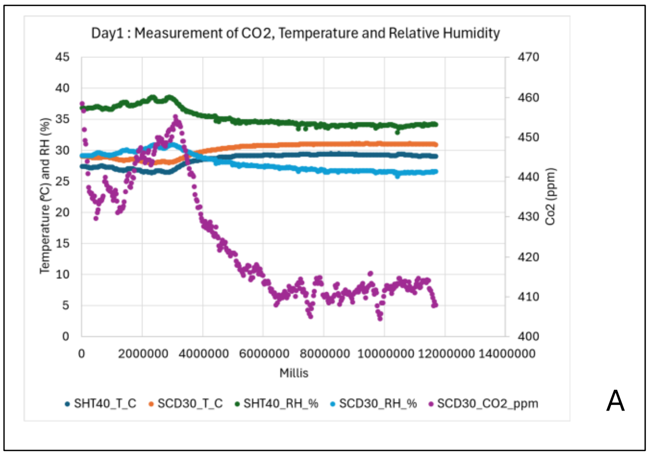
Results

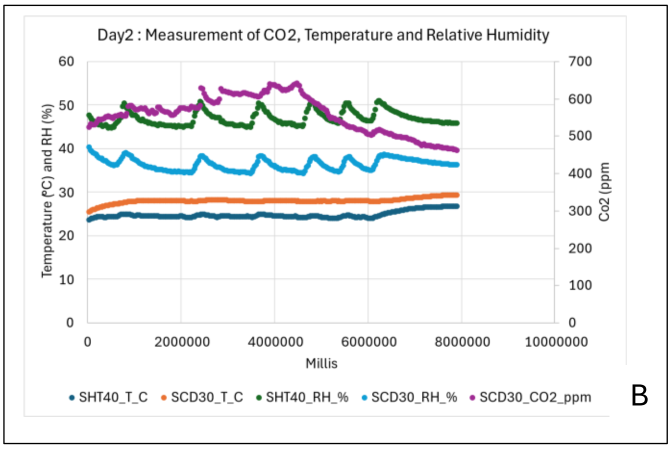
Plots of sensor measurements

The system was used to measure carbon dioxide, temperature, and relative humidity for three days. It should be noted that the whole day's duration was not considered (for example, from 9:00 am to 6:00 pm). This is because I could not leave the laptop supplying power for a long time. The maximum time duration was approximately 2 h when I was available in the office, and I ensured that the PC was operating to keep the system working.

The temperature and relative humidity values measured by the SHT40 and SCD30 sensors followed the same temporal patterns of change. The SCD30 sensor appeared to provide higher temperature values than the SHT40 sensor throughout the 3-day measurement; this observation reverses in terms of the RH values.

All the measurements were conducted during times when the air conditioner was either turned on or off. It was observed that CO_2 values increased when the air conditioner was in operation. When it was turned off, there was a sharp decline in the CO2 values. Although the intention was not to observe this behavior, it occurred. Figures 3 (A) and (B) show this trend.





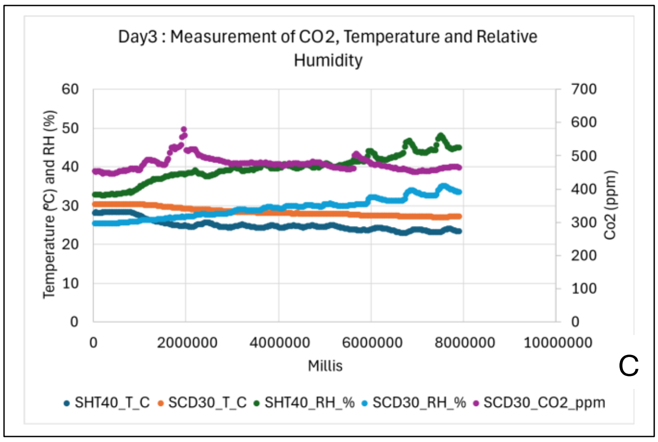


Figure 3 : (A); System was used to measure CO2, T/RH at day 1, (B) and (C) are measurements at day 2 and day 3, respectively.

SHT40_T_C - temperature measurement from the SHT40 T/RH sensor.

SCD30_T_C - temperature measurement from the SCD30 Co2 sensor

SHT40_RH_% and SCD30_RH_% are the relative humidity values obtained from both SHT40 and SCD30 sensors, respectively

SCD30_CO2_ppp is the CO2 values measured in parts per million

Conclusion

The project used an Arduino Uno microcontroller and temperature, relative humidity, and CO2 sensors to build a system for monitoring atmospheric parameters. The temperature, relative humidity, and CO2 were measured for three days to ensure system functionality. The sensors were sensitive to changes in room conditions, especially when the air conditioning was turned on and off, highlighting the importance of this observation for further investigation.